Inhomogeneous-wave beam tracing for lossy media

GOAD theory notes

1 Motivation and scope

GOAD currently handles absorption with three approximations. A beam transmitted into a lossy constituent takes its direction from a real effective-index Snell's law (the Macke 1996 port in `snell.rs`), its Fresnel coefficients from a complex $\cos\theta_t$ (`fresnel.rs`), and its attenuation from a scalar factor $\exp(-kn'd)$ applied along the propagation direction over the centroid-to-centroid distance d (`Field::propagate`). The beam cross-section keeps a uniform amplitude, and total internal reflection is decided by the real-index critical angle.

The rigorous picture is that the field inside a lossy constituent is an *inhomogeneous plane wave*: the planes of constant phase and the planes of constant amplitude have different normals. The definitive ray-level treatment is Chang, Walker and Hopcraft (2005), transcribed in `chang2005_absorbing_ray_tracing.md` and cited below as "Chang (N)". The IITM PGOM Fortran code implements the same physics in a polygon-beam tracer and serves as a reference implementation. This note derives the generalised beam tracing in GOAD's own notation (the beam tuple, the propagation factor T, and the aperture edge sum of `stable_fraunhofer_edge_sum.typ`), and proves that every generalised formula collapses to the current implementation in the lossless limit.

The scope is an absorbing particle in a lossless host medium. All interior interfaces may separate two lossy media; only the host is assumed transparent, which is what makes the far-field limit unambiguous.

2 Notation and the reduced wave vector

Material indices follow the reference paper: $\tilde n = n + n'i$ per constituent, wavenumber $k = 2\pi/\lambda$ in the host. Chang's apparent indices (N, K) are renamed (N, N') here, mirroring the (n, n') pairing, because K collides with the aperture-frame beam direction $\widehat{\boldsymbol{K}}$ of the far-field formulae.

A time-harmonic plane wave in a lossy medium has the complex wave vector $\boldsymbol{k}_c = k\big(N\hat{\boldsymbol{e}} + iN'\hat{\boldsymbol{f}}\big)$, where $\hat{\boldsymbol{e}}$ is the unit normal to planes of constant phase, $\hat{\boldsymbol{f}}$ the unit normal to planes of constant amplitude, and $N, N' \geq 0$ are the apparent refractive indices. Define the *reduced wave vector*

$$\boldsymbol{w} = \frac{\boldsymbol{k}_c}{k} = N\hat{\boldsymbol{e}} + i\boldsymbol{A}, \quad \boldsymbol{A} = N'\hat{\boldsymbol{f}}, \tag{1}$$

so that the field of a wave with amplitude E_0 at a reference point $\boldsymbol{r}_c$ is

$$\boldsymbol{E}(\boldsymbol{r}) = \boldsymbol{E}_0 \exp[ik\boldsymbol{w}\cdot(\boldsymbol{r}-\boldsymbol{r}_c)] = \boldsymbol{E}_0 \underbrace{\exp[ikN\hat{\boldsymbol{e}}\cdot(\boldsymbol{r}-\boldsymbol{r}_c)]}_{\text{phase}} \underbrace{\exp[-k\boldsymbol{A}\cdot(\boldsymbol{r}-\boldsymbol{r}_c)]}_{\text{amplitude}}. \tag{2}$$

The real vector $\boldsymbol{A}$ is the *absorption vector*: the gradient of the log-amplitude in units of k. Everything in this note is a statement about how $\boldsymbol{w}$ is created, transported, and consumed. The angle bookkeeping of Chang (the pairs θ, ψ, the inhomogeneity angle α, the azimuth ϕ between tangential directions) never needs to be materialised: it is all contained in the components of $\boldsymbol{w}$.

The beam tuple of the reference paper, $b = \big(F_b, \hat{\boldsymbol{k}}_b, \hat{\boldsymbol{e}}^b_\perp, \boldsymbol{S}_b, \phi_b, \tilde n_b\big)$, is extended by the wave state:

$$b = \big(F_b, \hat{\boldsymbol{k}}_b, \hat{\boldsymbol{e}}^b_\perp, \boldsymbol{S}_b, \phi_b, \tilde n_b, \boldsymbol{w}_b\big), \quad \hat{\boldsymbol{k}}_b = \operatorname{Re}\frac{\boldsymbol{w}_b}{|\operatorname{Re}\boldsymbol{w}_b|}. \tag{3}$$

Storing $\boldsymbol{w}_b$ (one complex 3-vector, or equivalently the real pair $N_b\hat{\boldsymbol{e}}_b$ and $\boldsymbol{A}_b$) is the entire cost of the generalisation at the state level. The propagation direction is no longer independent data; it is the direction of $\operatorname{Re}\boldsymbol{w}_b$.

3 The inhomogeneous plane wave

Substituting the plane-wave ansatz into the Helmholtz equation for a medium of index $\tilde{n}$ gives the single complex dispersion relation

$$\boldsymbol{w}\cdot\boldsymbol{w} = \tilde{n}^2. \tag{4}$$

Expanding with Equation 1, the real and imaginary parts are

$$N^2 - |\boldsymbol{A}|^2 = n^2 - n'^2, \quad N\,\hat{\boldsymbol{e}}\cdot\boldsymbol{A} = nn', \tag{5}$$

which are Chang (3) with $\boldsymbol{A} = N'\hat{\boldsymbol{f}}$ and $\hat{\boldsymbol{e}}\cdot\hat{\boldsymbol{f}} = \cos\alpha$. Two consequences used throughout:

1. The attenuation per unit length *along the ray* is $k\,\boldsymbol{A}\cdot\hat{\boldsymbol{e}} = knn'/N$, which is Chang (15). It is always at most kn', since $N \geq n$: the standard method overestimates path loss.
2. In a lossless medium ($n' = 0$) either $\boldsymbol{A} = 0$ (homogeneous wave) or $\boldsymbol{A} \perp \hat{\boldsymbol{e}}$ with $|\boldsymbol{A}|^2 = N^2 - n^2$ (a transversely decaying wave; the evanescent wave is the limiting case with N below the tangential invariant). Both occur in GOAD: the incident illumination is the first, a beam exiting an absorbing particle is generically the second.

A homogeneous wave in a lossy medium is the special case $\hat{\boldsymbol{f}} = \hat{\boldsymbol{e}}$, for which Equation 5 forces $N = n$, $N' = n'$ and Equation 2 reduces to the current beam field with $\boldsymbol{w} = \tilde{n}\hat{\boldsymbol{k}}$. The current implementation is therefore the general formalism with $\boldsymbol{w}_b$ *forced* to $\tilde{n}_b\hat{\boldsymbol{k}}_b$ for every beam.

4 Interface transfer

Let $\hat{\boldsymbol{n}}$ be the interface normal oriented from medium 1 into medium 2, and split any vector into tangential and normal parts, $\boldsymbol{w} = \boldsymbol{w}_t + w_n\hat{\boldsymbol{n}}$. Matching the total field phase and amplitude over the interface plane (Chang (8), applied to the full complex vector) gives the entire generalised Snell's law in one line:

$$\boldsymbol{w}_{t,r} = \boldsymbol{w}_{t,2} = \boldsymbol{w}_{t,1} =: \boldsymbol{w}_t. \tag{6}$$

The normal component of the transmitted wave follows from Equation 4 in medium 2:

$$\boldsymbol{w}_2 = \boldsymbol{w}_t + \gamma\hat{\boldsymbol{n}}, \quad \gamma = \sqrt{\tilde{n}_2^2 - \boldsymbol{w}_t\cdot\boldsymbol{w}_t}, \tag{7}$$

with the principal branch of the square root, which has $\operatorname{Re}\gamma \geq 0$ (energy proceeds into medium 2) and returns $+i\sqrt{|\cdot|}$ on the branch cut (evanescent decay into medium 2). The reflected wave stays in medium 1 with its normal component negated:

$$\boldsymbol{w}_r = \boldsymbol{w}_t - w_{n,1}\hat{\boldsymbol{n}}. \tag{8}$$

Note that Equation 8 mirrors the absorption vector as well as the propagation direction; the reflected beam inherits the incident inhomogeneity, which is Chang (12) for $\hat{\boldsymbol{f}}_r$.

4.1 Equivalence with Chang's closed form

Writing $\boldsymbol{w}_t = N_s\hat{\boldsymbol{t}}_e + iK_s\hat{\boldsymbol{t}}_f$ with $N_s = N_1\sin\theta_i$, $K_s = N_1'\sin\psi_i$ and $\hat{\boldsymbol{t}}_e\cdot\hat{\boldsymbol{t}}_f = \cos\phi$, the argument of the root in Equation 7 is

$$\gamma^2 = \underbrace{\left(n_2^2 - n_{2'}^2\right)}_{M} - N_s^2 + K_s^2 + 2i\underbrace{(n_2 n_{2'} - N_s K_s \cos\phi)}_{P}. \tag{9}$$

Splitting $\gamma = \gamma_r + i\gamma_i$ and using $N_2^2 = |\mathrm{Re}\,\boldsymbol{w}_2|^2 = N_s^2 + \gamma_r^2$ gives

$$N_2^2 = \frac{1}{2}\left[N_s^2 + K_s^2 + M + \sqrt{\left(N_s^2 - K_s^2 - M\right)^2 + 4P^2}\right], \quad N_{2'}^2 = N_2^2 - M, \tag{10}$$

which is the positive root of the quartic Chang (10) and the closed form implemented in PGOM (`permittivity.f90`). The transmitted directions $\hat{\boldsymbol{e}}_2 = \mathrm{Re}\,\frac{\boldsymbol{w}_2}{N_2}$ and $\hat{\boldsymbol{f}}_2 = \mathrm{Im}\,\frac{\boldsymbol{w}_2}{N_{2'}}$ reproduce Chang (12). None of these intermediates are needed by an implementation that carries $\boldsymbol{w}$; Equation 7 is one complex square root and one vector sum.

4.2 Special cases

First entry. The incident illumination is homogeneous in a lossless host: $\boldsymbol{w}_1 = n_{\mathrm{med}}\hat{\boldsymbol{k}}_0$, so $\boldsymbol{w}_t$ is real, $K_s = 0$, and Equation 10 reduces to the Liou form used at the first interface in PGOM. The transmitted absorption vector is $\boldsymbol{A}_2 = \gamma_i\hat{\boldsymbol{n}}$: attenuation points *into the surface*, not along the refracted ray. This is the single largest qualitative difference from the current code, which sets $\boldsymbol{A} = n'\hat{\boldsymbol{e}}$ implicitly.

Interior interfaces. The incident wave is already inhomogeneous; Equation 6 and Equation 7 apply unchanged. This is where the current complex-Snell treatment breaks down (see Section 5) and where Chang's quartic is genuinely needed.

Exit into the lossless host. $\tilde{n}_2 = n_{\mathrm{med}}$ real, but $\boldsymbol{w}_t$ is complex because the interior wave carries attenuation. By Equation 5 with $n' = 0$, the transmitted wave has $\boldsymbol{A}_2 \perp \hat{\boldsymbol{e}}_2$: a propagating wave with a purely transverse amplitude ramp. This ramp is the exact boundary field that the far-field stage must diffract (Section 8); it is not an artefact but the physical imprint of the oblique absorbing path on the exiting wavefront.

Total internal reflection. From Equation 10, $N_2 \geq N_s$ always, with equality only when $P = 0$ and $N_s^2 \geq K_s^2 + M$. For a lossy medium 2 (or an inhomogeneous incident wave with $P \neq 0$) the inequality is strict, $\sin\theta_{t,2} = N_s/N_2 < 1$, and a transmitted wave always exists: *there is no total internal reflection in lossy media* (Chang section 4). The classical TIR emerges as the lossless limit: $n_2' = 0$, $K_s = 0$, $N_s > n_2$ gives $\gamma^2 < 0$, $\gamma = i\sqrt{N_s^2 - n_2^2}$, a pure evanescent wave carrying no normal flux, and $|r| = 1$ from the Fresnel formulae below. The beam-level consequence: the decision to spawn a transmitted beam becomes an energy-threshold decision, not an angular one. The current branches `theta_i > (n2.re/n1.re).asin()` in `beam.rs` should be replaced by the general transfer, with the lossless evanescent case suppressed by the existing power threshold (its transmitted flux is exactly zero) or by an explicit $\mathrm{Re}\,\gamma = 0$ test for determinism.

5 Generalised Fresnel coefficients

Define the complex normal components $\kappa = \boldsymbol{w}\cdot\hat{\boldsymbol{n}}$ for the incident and transmitted waves,

$$\kappa_i = \boldsymbol{w}_1\cdot\hat{\boldsymbol{n}} = N_1\cos\theta_i + iN_1'\cos\psi_i, \quad \kappa_t = \gamma. \tag{11}$$

Chang (14) gives the reflection and transmission coefficients entirely in terms of these:

$$r_s = \frac{\kappa_i - \kappa_t}{\kappa_i + \kappa_t}, \quad t_s = \frac{2\kappa_i}{\kappa_i + \kappa_t}, \quad r_p = \frac{\tilde{n}_2^2\kappa_i - \tilde{n}_1^2\kappa_t}{\tilde{n}_2^2\kappa_i + \tilde{n}_1^2\kappa_t}, \quad t_p = \frac{2\tilde{n}_1\tilde{n}_2\kappa_i}{\tilde{n}_2^2\kappa_i + \tilde{n}_1^2\kappa_t}. \tag{12}$$

5.1 Exact reduction to the current implementation

For a *homogeneous* incident wave, $\boldsymbol{w}_1 = \tilde{n}_1 \hat{\boldsymbol{e}}_1$, so $\kappa_i = \tilde{n}_1 \cos\theta_i$ and $\boldsymbol{w}_t \cdot \boldsymbol{w}_t = \tilde{n}_1^2 \sin^2\theta_i$; then

$$\kappa_t = \gamma = \sqrt{\tilde{n}_2^2 - \tilde{n}_1^2 \sin^2\theta_i} = \tilde{n}_2 \cos\theta_t^c, \tag{13}$$

where $\cos\theta_t^c$ is precisely the complex cosine computed by `fresnel.rs::cos_theta_t`. Substituting into Equation 12 and cancelling $\tilde{n}_1\tilde{n}_2$ from r_p recovers the four formulae of the reference paper's Appendix A term for term. Two corollaries:

1. The current Fresnel treatment is *exact* at every interface whose incident wave is homogeneous, which includes the first entry into an absorbing particle, even between two lossy media. The generalisation changes nothing there.
2. The error of the current treatment is confined to interior interfaces reached by beams that are already inhomogeneous. There, replacing $\tilde{n}_2 \cos\theta_t^c$ by the true γ of Equation 7, and $\tilde{n}_1 \cos\theta_i$ by the true κ_i, is the entire correction: the algebraic form of Equation 12 is unchanged.

5.2 Polarisation basis

The exact field decomposition for inhomogeneous waves uses the complex orthonormal pair of Chang (13), $\boldsymbol{h}_1 = \hat{\boldsymbol{n}} \times \boldsymbol{w} / \sqrt{(\hat{\boldsymbol{n}} \times \boldsymbol{w}) \cdot (\hat{\boldsymbol{n}} \times \boldsymbol{w})}$ and $\boldsymbol{h}_2 = \boldsymbol{w} \times \boldsymbol{h}_1 / \tilde{n}$, which are *bivectors* (complex-valued vectors) whenever the tangential phase and attenuation directions differ. The coefficients Equation 12 are the scalars that act on the $(\boldsymbol{h}_1, \boldsymbol{h}_2)$ components.

GOAD's field structure assumes a real basis $\left(\hat{\boldsymbol{e}}_\perp, \hat{\boldsymbol{e}}_\parallel\right)$. Since $\hat{\boldsymbol{n}} \times \boldsymbol{w} = \hat{\boldsymbol{n}} \times \boldsymbol{w}_t$, the basis $\boldsymbol{h}_1$ is real exactly when the tangential parts of $\hat{\boldsymbol{e}}$ and $\hat{\boldsymbol{f}}$ are parallel; in general the real-basis description misses a component of relative size $O(|\boldsymbol{A}|\ /N)$, which is $O(n'/n)$. There are two consistent options:

1. *Exact*: promote $\hat{\boldsymbol{e}}_\perp$ (and the basis-rotation matrices) to complex vectors, as PGOM does (`VA`, `VB`, `VK` are complex). The rotation matrix construction of `Field::rotation_matrix` continues to apply with complex dot products.
2. *Real-basis approximation*: keep real bases built from $\hat{\boldsymbol{e}}$, apply the scalar coefficients Equation 12. The neglected terms are first order in n'/n, the same order as the Fresnel phase corrections themselves, so this option captures the dominant correction (the modified κ values) while leaving a residual of the same formal order with a smaller prefactor. Whether the prefactor is actually small must be established numerically in the validation phase, not assumed.

The theory is recorded here for both; the choice is an implementation decision gated on the Phase B measurements.

6 Beam propagation, splitting, and the amplitude ramp

6.1 Propagation between interfaces

The current `Field::propagate(dist, k, n)` winds phase by knd and multiplies amplitude by $\exp(-kn'd)$, with d the scalar displacement along $\hat{\boldsymbol{k}}_b$. The general rule, from Equation 2, uses the vector displacement $\boldsymbol{d} = \boldsymbol{r}_{c'} - \boldsymbol{r}_c$ between the parent centroid and the child centroid:

$$\phi_b \to \phi_b + kN\,\hat{\boldsymbol{e}} \cdot \boldsymbol{d}, \quad \boldsymbol{S}_b \to \boldsymbol{S}_b \exp(-k\boldsymbol{A} \cdot \boldsymbol{d}). \tag{14}$$

The phase term is unchanged in structure (the scalar d used today *is* $\hat{\boldsymbol{e}} \cdot \boldsymbol{d}$). The amplitude term is not: for an inhomogeneous wave $\boldsymbol{A} \cdot \boldsymbol{d} \neq (\boldsymbol{A} \cdot \hat{\boldsymbol{e}})(\hat{\boldsymbol{e}} \cdot \boldsymbol{d})$, so `propagate` must receive the full displacement vector rather than a scalar distance. The same applies to the phase re-referencing in `earcut` and `remainders_to_beams`.

Like the accumulated phase, the accumulated log-amplitude is a running sum of increments; unlike the phase, its consumer is an exponential rather than a trigonometric function, so it needs no modular reduction and f32 relative error does not grow with the total. Carrying it as a separate f64 accumulator alongside `phase` is nevertheless cheap and removes any doubt; decide at implementation time.

6.2 The ramp across a beam

Because $\boldsymbol{S}_b$ is defined at the centroid and the field over F_b is Equation 2, the amplitude across the beam polygon varies as $\exp(-k\boldsymbol{A}_t \cdot (\boldsymbol{r} - \boldsymbol{r}_c))$, with $\boldsymbol{A}_t$ the projection of $\boldsymbol{A}$ into the plane of F_b. No per-vertex state is required: PGOM's per-vertex attenuation arrays (`TP`) are recoverable at any time as $k\boldsymbol{A} \cdot \big(\boldsymbol{r}_j - \boldsymbol{r}_c\big)$. GOAD should carry only $\boldsymbol{w}_b$ and evaluate vertex exponents lazily where they are consumed (the effective cross-section and the far-field integral).

PGOM's facet update rule for its absorption vector, $\boldsymbol{A} \leftarrow \boldsymbol{A} + (N_i - \boldsymbol{A} \cdot \hat{\boldsymbol{e}})\hat{\boldsymbol{n}}/(\hat{\boldsymbol{e}} \cdot \hat{\boldsymbol{n}})$, is the combination "keep the tangential part, restore the dispersion constraint $\boldsymbol{A} \cdot \hat{\boldsymbol{e}} = nn'/N$ by adjusting the normal part". This is algebraically identical to the transfer Equation 7: both determine $\boldsymbol{A}_2$ from tangential continuity plus Equation 5. The two derivations agreeing is a useful cross-validation of Equation 7.

6.3 Splitting

When a beam is clipped into sub-polygons or earcut into simple faces, each child re-references $\boldsymbol{S}$ and ϕ to its own centroid via Equation 14 with $\boldsymbol{d}$ the centroid offset. The ramp is then automatically consistent: parent and child describe the same field where they overlap, because both are restrictions of the same inhomogeneous plane wave. This is the same invariant the current code maintains for phase, extended to amplitude.

7 Power, effective cross-section, and the absorption ledger

7.1 Energy flux

For the TE component the time-averaged Poynting vector of an inhomogeneous wave is exactly

$$\langle \boldsymbol{S} \rangle = \frac{1}{2}\sqrt{\frac{\varepsilon_0}{\mu_0}}\, N\, |E|^2\, e^{-2k\boldsymbol{A}\cdot\boldsymbol{r}}\, \hat{\boldsymbol{e}}, \tag{15}$$

that is, flux along $\hat{\boldsymbol{e}}$ with the *apparent* index N as the flux factor, not n. For the TM component there is an additional term proportional to $\mathrm{Im}[\boldsymbol{E}^*(\boldsymbol{E}\cdot\boldsymbol{A})]$, of relative size $O(|\boldsymbol{A}|\ /N)$, which tilts the flux slightly out of $\hat{\boldsymbol{e}}$. The tracer adopts Equation 15 for both polarisations and the TM correction joins the error budget (Section 10). The current `Beam::power()` uses `refr_index.re`, i.e. n; the generalisation replaces it by $N = |\mathrm{Re}\,\boldsymbol{w}|$.

7.2 Effective cross-section

The power a beam carries through its polygon is the flux integrated over the ramp:

$$P_b = \frac{1}{2} N_b \,\|\boldsymbol{S}_b\|^2\, |\hat{\boldsymbol{e}}_b \cdot \hat{\boldsymbol{n}}_{F_b}| \iint_{F_b} e^{-2k\boldsymbol{A}_t\cdot(\boldsymbol{r}-\boldsymbol{r}_c)}\, \mathrm{d}A, \tag{16}$$

which generalises `csa()` (the integral reduces to the polygon area when $\boldsymbol{A}_t = 0$). The integral is the Fourier transform of the polygon indicator at a purely imaginary argument: in the aperture frame with in-plane ramp components $(\mu, \nu) = 2k\big(A_x, A_y\big)$,

$$\iint_{F_b} e^{-(\mu x+\nu y)}\,\mathrm{d}x\,\mathrm{d}y = A(i\mu, i\nu), \tag{17}$$

with $A(p,q)$ the edge-sum transform of `stable_fraunhofer_edge_sum.typ` continued to complex arguments (Section 8). One evaluator therefore serves both the far field and the power bookkeeping; PGOM's separate `PYEXP` contour form for D_{eff} is this same identity.

The beam-queue prioritisation and the power threshold should use Equation 16, otherwise a strongly ramped beam is ranked by an amplitude its bright edge does not represent.

7.3 Absorption ledger

The power absorbed along one propagation leg is the difference between Equation 16 evaluated at the start and at the end of the leg (the amplitude factors differ, and for oblique legs the ramp differs too). This replaces the current midpoint tally `absorbed_intensity * area * cos(theta_i) * n1.re` in `intersections_to_beams`, which is the $\boldsymbol{A} = n'\hat{\boldsymbol{e}}$, uniform-ramp special case. The ledger sums to the absorption cross-section; extinction continues to come from the optical theorem on the forward amplitude, which is unaffected because the host is lossless, and energy-conservation checks compare the two as today.

8 Far field with a ramped aperture

An exiting beam is a propagating wave in the host with a purely transverse absorption vector (Equation 5 with $n' = 0$). Its boundary field on the aperture polygon is the plane wave of the current treatment multiplied by the real exponential ramp $\exp(-k\boldsymbol{A}_t \cdot (\boldsymbol{r} - \boldsymbol{r}_c))$.

8.1 Analytic continuation of the edge sum

In the aperture frame the Fraunhofer integral becomes

$$A_U(p,q) = \iint_A e^{i(px+qy)} e^{-(\mu x+\nu y)}\,\mathrm{d}x\,\mathrm{d}y = A(p+i\mu, q+i\nu), \tag{18}$$

with $(\mu,\nu) = k\big(A_x, A_y\big)$ the in-plane ramp components. The stable edge sum evaluates A at complex arguments with no structural change: the per-edge identity $e^{i\delta_j}\varphi\big(\omega_j\big) = \big(e^{i\delta_{j+1}} - e^{i\delta_j}\big)/\big(i\omega_j\big)$ is an identity of entire functions and holds for complex δ, ω; the vertex factors become $e^{i\delta_j^r} e^{-\delta_j^i}$, one `cis` and one real exponential per vertex; and φ keeps its Taylor branch with the threshold applied to $|\omega|$ (complex modulus). In the code, the continuation enters through the wavevector mismatch: `kxx` and `kyy` gain imaginary parts kA_x, kA_y, i.e. the mismatch is computed from $\boldsymbol{w}$ of the exiting beam rather than from $\tilde{n}\hat{\boldsymbol{k}}$.

8.2 Stability at complex argument

The three error mechanisms of the real-argument analysis carry over with one addition:

1. *Difference-form cancellation.* Unchanged in form: relative error $\sim \frac{\varepsilon_{\mathrm{f32}}}{|}\omega|$ near $\omega = 0$; the existing threshold $|\omega| < 0.3$ transfers with $|\omega|$ the complex modulus, since the Taylor truncation bound $|\omega\frac{|^6}{5040}$ is a bound in modulus.
2. *Small-argument limit.* The moment expansion $A = S + i\big(pM_x + qM_y\big) + O\big((\kappa R)^2\big)$ is an expansion of an entire function and holds verbatim for complex p, q with $\kappa = \max(|p|, |q|)$.
3. *Dynamic range (new).* The vertex weights $e^{-\delta_j^i}$ span a ratio $\exp(2k\ |\boldsymbol{A}_t|\ r_{\max})$ across the polygon. The edge sum then computes a result that can be dominated by the bright side while f32 rounding is set by the largest terms; relative error grows linearly with the dynamic range ratio in the worst case. Physically a large ratio means the dim side of the beam is negligible, so the absolute error

remains controlled, but this must be measured, not argued: the Phase B harness sweeps $k\ |\boldsymbol{A}_t|\ r_{\max}$ from 0 to the largest value reachable (bounded by the total attenuation a beam can suffer before the power threshold kills it, roughly $-\ln(\mathrm{P}_{\mathrm{th}})$ in exponent units).

8.3 Field of view

The FOV cone half-angle currently scales with $\lambda/r_{\max}$. A strong ramp concentrates the effective aperture into a region smaller than $r_{\max}$, widening the true diffraction lobe. A conservative generalisation replaces $r_{\max}$ by an effective radius $r_{\mathrm{eff}} = \sqrt{A(i\mu, i\nu)/\pi}$, monotonically shrinking with the ramp. Whether the extra `polygon_ft` evaluation pays for itself is a Phase D question; the conservative interim choice is to leave the FOV cone at its ramp-free width, which can only over-include bins.

8.4 Polarisation matrix

The Karczewski matrix is derived for a homogeneous plane wave through the aperture. With a ramp, the aperture magnetic field acquires a term $\propto \boldsymbol{A} \times \boldsymbol{E}$, out of phase with the electric surface current, of relative size $|\boldsymbol{A}|\ /N$. Since the Karczewski entries are algebraic in the components of $\widehat{\boldsymbol{K}}$, the natural exact treatment is the same analytic continuation as everywhere else: evaluate them at the complex direction $\boldsymbol{w}/n_{\mathrm{med}}$ of the exiting wave. The interim approximation keeps the real $\widehat{\boldsymbol{K}} = \hat{\boldsymbol{e}}$ and accepts an $O(|\boldsymbol{A}|\ /N)$ error in the polarisation factor only (the scalar diffraction factor is exact either way). Phase B validates the continued form against direct quadrature of the vector integral before Phase D commits to either.

9 Collapse to the current physics

Setting $n' = 0$ for every constituent (and the host):

1. *State.* Equation 7 preserves realness: real $\boldsymbol{w}_t$, real $\tilde{n}_2^2$, and below the critical angle real γ, so $\boldsymbol{A} = 0$ for every beam and $\boldsymbol{w} = n\hat{\boldsymbol{k}}$ identically. Above the critical angle γ is purely imaginary, the transmitted flux Equation 15 is zero, and no transmitted beam survives the power threshold: classical TIR, including the $|r| = 1$ phase-only reflection of the current code.
2. *Snell.* Equation 7 with real arguments is $\sin\theta_t = n_1 \sin\theta_i / n_2$; `get_theta_t` (Macke) is the homogeneous special case of Equation 10 and agrees.
3. *Fresnel.* Equation 13 shows Equation 12 equals the current formulae exactly whenever the incident wave is homogeneous, which with $\boldsymbol{A} = 0$ is every interface.
4. *Propagation.* Equation 14 with $\boldsymbol{A} = 0$: amplitude factor 1, phase increment $kn\hat{\boldsymbol{e}} \cdot \boldsymbol{d}$, the current winding.
5. *Power.* $N = n$, the ramp integral in Equation 16 is the polygon area: `power()` and `csa()` unchanged.
6. *Far field.* $(\mu, \nu) = 0$: Equation 18 is the real edge sum, bit-identical code path if the complex evaluator short-circuits on a zero imaginary part, or tolerance-identical if it does not.

For absorbing media under the current approximation ($\boldsymbol{w}$ forced to $\tilde{n}\hat{\boldsymbol{k}}$), the general formulae reproduce the current behaviour exactly as well, which provides a second regression axis: the new code with forced homogeneity must match the old code on the absorbing benchmarks before the forcing is removed.

10 Error budget and applicability

From Chang section 3, with $x = n'/n$:

- Refraction angles and path loss: corrections are $O(x^2)$ (Chang (16), (17)). Negligible below $x \sim 0.05$.

- Fresnel amplitudes: corrections are $O(x)$ (Chang (18)), and the induced ellipticity (the S_{34}-type elements) is directly proportional to x. This is the leading physical effect and the reason polarisation output is the most sensitive validation target.
- Reflectivities: $O(x^2)$ but with a large prefactor $4n^3/(n^2-1)^2$ (Chang (19)); at $n = 1.5$ a 1 percent deviation is reached already at $x \sim 0.034$.
- No-TIR transmission: qualitatively absent from the current code; the transmitted power in the former TIR zone is first order in x and matters for beams trapped by repeated internal reflection.

Applicability guideline (Chang section 6): the current physics is adequate for unpolarised quantities below $x \sim 0.07$ and for polarised quantities below $x \sim 0.01$. The GOAD absorbing benchmarks at $\tilde{n} = 1.31 + 0.1i$ sit at $x = 0.076$, outside both.

f32 concerns specific to this rework: the accumulated log-amplitude (bounded by $-\ln \mathrm{P}_{\mathrm{th}}$, no precision issue), the ramp dynamic range in the edge sum (measured in Phase B), and γ near the branch point (a `sqrt` of a small complex number, benign in f32 but the regression tests should pin the near-critical behaviour).

11 Implementation implications

Deferred to `lossy_media_plan.md` Phase D; the state-level summary is:

- `Beam`/`Field` gain $\boldsymbol{w}$ (equivalently $\boldsymbol{A}$ plus N); `refr_index` remains the material index for Fresnel.
- `snell.rs` shrinks to Equation 7; `fresnel.rs` keeps its algebraic form with κ inputs from $\boldsymbol{w}$.
- `propagate`, `earcut`, `remainders_to_beams` take vector displacements.
- `power()`, `csa()`, the queue ordering, and the absorbed-power tally move to N and the ramp integral Equation 16.
- `polygon_ft` continues to complex (p, q); `EdgeData` unchanged; the mismatch `kxx`, `kyy` becomes complex, sourced from $\boldsymbol{w}$.
- The polarisation basis decision (complex bivector versus real approximation) is the one genuinely open design choice, gated on Phase B numbers.